

MODULE 1 · THE INTEGRATION FOUNDATION

The Integration Backbone

Week 1 — the single source of truth, and why a backbone that works can still fail to pay.

Friday, September 4, 2026 · 8:00–10:30 AM · 2 MTC Room 802

We build the floor together, then start your project

8:00

Overview + why this course, now

The market, the arc, the three threads, the four companies, and the capstone you build all term

8:20

The floor

What a backbone is, the shared data model, control vs decision points — the ERP fundamentals, from the ground up

8:43

Read a backbone

Two placement questions, the leverage gap, and two real cases — NYU's own library and Lidl

9:20

Your team kicks off

Meet your team, pick a candidate company, and run the diagnostic — the first step of your capstone

10:00

Prime Week 2 + homework

Why a working backbone still fails to pay — and the weekly reading rhythm begins

You are surrounded by integrated systems

Scanning a QR code on a Citi Bike dock isn't an app-to-device ping — it's a live integration across hardware, finance, and logistics. Three layers execute in milliseconds so you never think about it:

THE SPINE

Asset database

Tracks 30,000+ bikes, dock locks, battery levels, and subscriptions in real time — one unified data backbone.

THE ENGINE

Orchestration

Authorizes the card hold, fires the physical lock release, and updates the dock display — all in the moment you scan.

THE BRAIN

Rebalancing agent

Senses empty docks, plans box-truck routes, awards Bike Angel points, and reroutes logistics — in real time.

Integrate → automate → make agentic. The same three layers that run a bike dock run the enterprise — and the whole course climbs them in order. You cannot automate what you have not integrated.

The gap between investment and outcome is where your career lives

\$179K

median pay for enterprise Integration Architects — the role this course trains you for.

Glassdoor, 2025

60%

of executives cite scarce tech & integration talent as the #1 constraint on transformation.

McKinsey, 2025

56%

of Chief Supply Chain Officers say integrating AI with legacy systems is a major challenge.

Gartner, 2026

~40%

of companies report any enterprise-level EBIT from AI — pilots rarely reach the backbone.

McKinsey, 2026

The skill is scarce, well paid, and exactly what you'll practise here. It isn't using AI — it's turning technology into business value and leading the change. You leave with that method, and a board-ready case on a real company that proves you can do it.

Three threads, running through every company

CUSTOMER

“Are we serving customers reliably?”

Order fulfillment, service quality, experience.

OPERATIONS & SUPPLY CHAIN

“Are we synchronized — or firefighting?”

Planning and sync so the promise can be met.

CASH MANAGEMENT

“Is cash moving fast enough to keep us alive?”

Conversion speed, working capital. A leading reason businesses fail.

FOUR COMPANIES we carry all term — every lesson asked of each:

Global CPG

on SAP

Mid-market manufacturer

on NetSuite

Series-B startup

Stripe · Shopify · QuickBooks

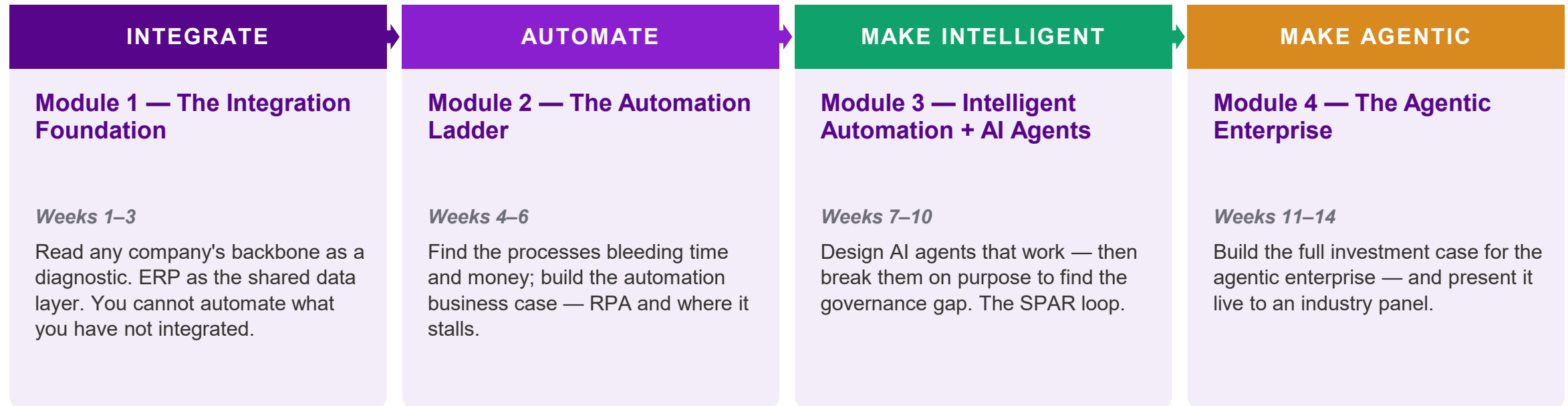
50-person services firm

HubSpot · Notion

These aren't abstract concepts — they're what your future employer is losing sleep over. Ask each thread of a Fortune-500 and a seed-stage startup in the same breath, and every question in this course gets sharper.

Four modules climb one arc

The course is one climb — integrate, then automate, then make intelligent, then make agentic. Each module is one rung, and each builds a piece of your final board case:



You are here: Module 1, the foundation. Everything above only pays off if the layer below is solid — that's why we integrate first, and why the labs and your board memo build in the same order.

The note teaches. The room applies.

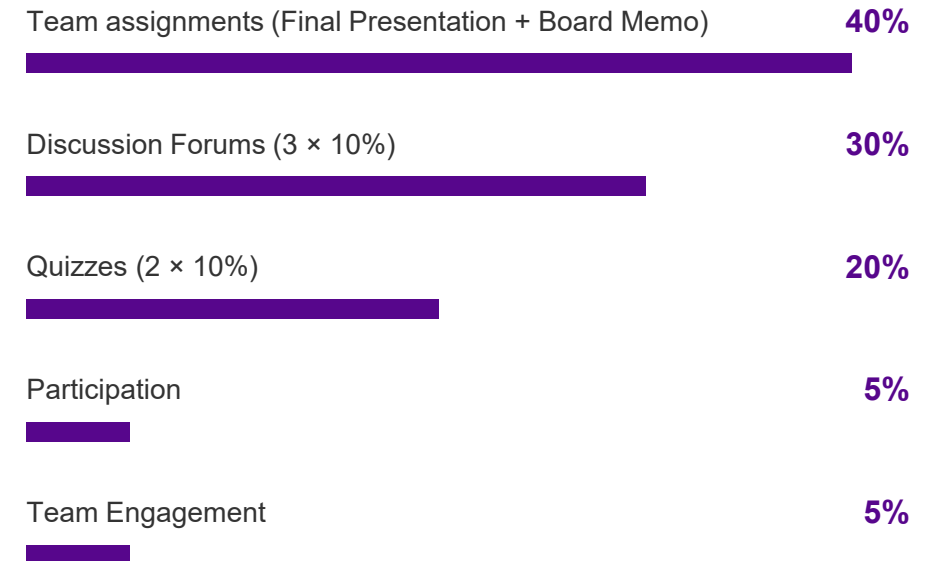
- Each week has a study note that teaches the concepts. You read it before that week's class.
- Class is a working session that activates and applies the note — we don't re-read it in the room.
- Each week's homework builds one defined piece of your final board presentation.
- That weekly rhythm begins with Week 2 — the note you receive today, with a full week's lead. Today we build the floor together in the room.

Everything you need today, we build here. Come each week having read the note, and class becomes the place you apply it to your own company — and your teammates'.

14 Fridays, teams, labs — and what it's worth

- 14 Fridays, 8:00–10:30, Sep 4 – Dec 11 (no class Thanksgiving).
- Teams of 4–5, already assigned; you meet yours today and build your project together.
- Four hands-on labs (Weeks 3, 6, 9, 12) on a no-code stack — no programming required.
- Three stack paths — Default (Airtable/Sheets + Gemini), Enterprise (AWS), or Custom. The Default earns full marks.
- You work with a real organisation; all project data is synthetic, so nothing confidential is ever used.

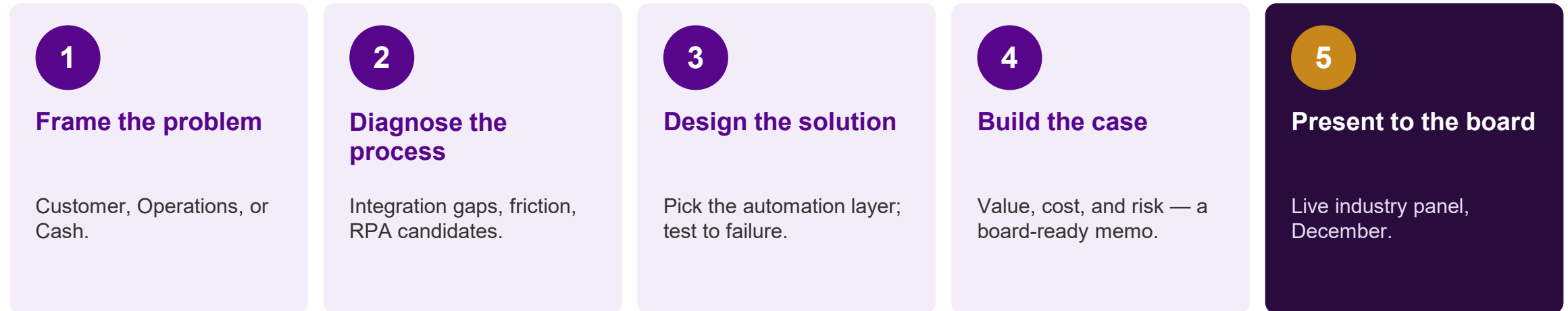
GRADE BREAKDOWN



Who this is for: strategic enablement and change leadership. You'll translate business goals into system requirements, manage change, and evaluate the ROI of digital investment — the capabilities employers pay for, practised on a real company.

A methodology you can charge for

One repeatable method — the same five moves, applied to a real company, presented to an industry panel in December:



This is not a technology course. It's a course in making technology deliver what it was bought to deliver. By December you have the method and a portfolio piece that proves it — exactly what the market rewards. You start move 1 today, in your team.

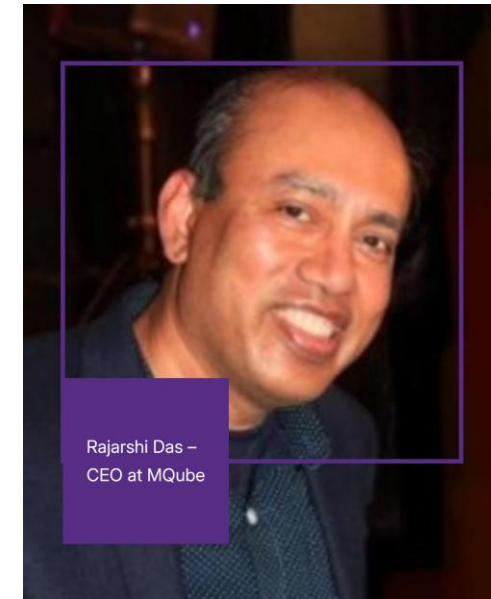
One board presentation — each week fills a slide

- Every one of the 14 weeks maps to a defined slide of your final board presentation.
- You receive the capstone storyboard today and map your team's deck to it.
- This week fills S1 (title & company) and S3 (context) — and feeds Board Memo Section 2, Context & Background.
- Your homework is that seed: a backbone-context note — the stack, the key processes, the archetype, the primary thread.

Choosing what earns a board slide is itself the discipline. You're deciding, week by week, what a board actually needs to see. Treat today's cases — especially NYU's own library — as worked examples of exactly that.

Who you'll learn from — and present to

Practitioners from industry join us through the term to guest-lecture — and they sit on the Executive Panel your team presents its final board case to in December.





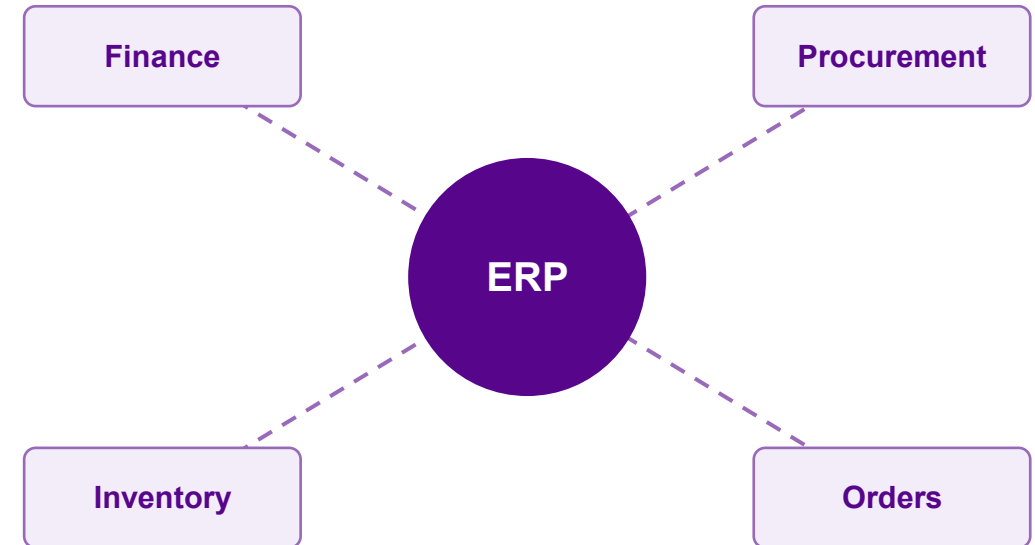
NOW THE FLOOR

First: what a backbone actually is

The next twenty minutes build the ERP fundamentals from the ground up — everything the rest of the day stands on.

Every company runs on a nervous system

- An ERP is the shared data backbone connecting every function — finance, procurement, inventory, production, customer orders.
- When it works, a single sale automatically triggers procurement, adjusts stock, and moves cash. No re-keying.
- When it doesn't, people re-enter data between disconnected systems — orders go missing, cash stalls, and automation amplifies the dysfunction instead of fixing it.



Most companies' nervous systems are broken. Systems integration is the discipline of diagnosing what's broken and sequencing the fix — before any automation is layered on top.

“ERP” is not one thing

A backbone holds four broad zones — leave one out and the company has a blind spot exactly there:

Financial

Human-capital

Operations & logistics

Commercial

And backbones differ along three market dimensions — cutting across the four archetypes' size scale:

Deployment

On-premise, cloud, or hybrid.

Scope

Generalist / horizontal vs industry-specific / vertical.

Market tier

Tier I (large global), II (mid-market), III (small business).

Reading a backbone means placing it on these lines — not only on the size scale. Is your company cloud or on-prem, generalist or vertical, which tier? Job ads and vendor pages give most of it away.

One fact, one place — and the process that guards it

- Strip the backbone to its defining property: a single source of truth. For any fact — an order quantity, an invoice status, a stock level — exactly one authoritative record everyone reads from and writes to.
- The shared data model makes it possible: “customer 4471” means the same entity to finance and to fulfillment.
- When it's absent, every department keeps its own version of reality, and reconciliation — making the copies agree — becomes the organisation's hidden full-time job.
- A real backbone also enforces the process: it won't post an invoice against an unapproved order, or ship goods never reserved.

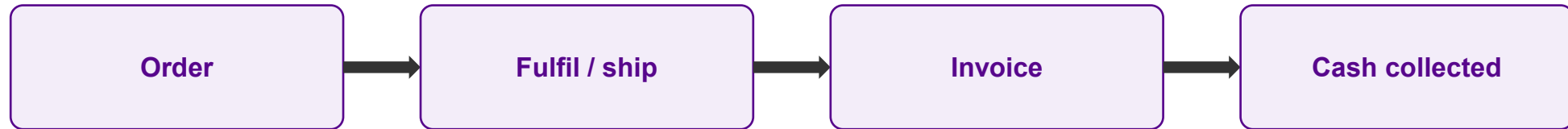
When it's absent...

every department quietly keeps its own version of reality, and the endless work of making the copies agree becomes the hidden full-time job.

Single source of truth and process enforcer are inseparable. The truth is trustworthy only if the steps that created it were followed in order. That's why the backbone is both at once.

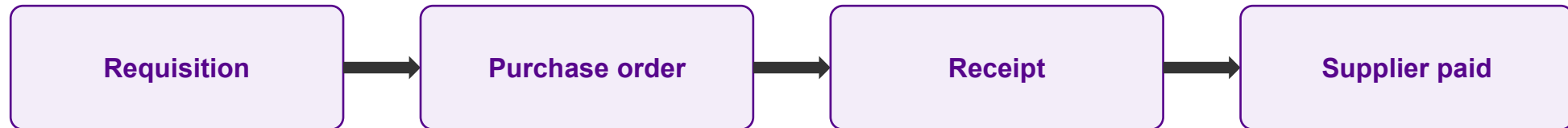
Two flows run through every integration decision

O2C



Order-to-Cash — a customer's order, all the way to cash in the door.

P2P



Procure-to-Pay — a requisition, all the way to a supplier paid. (Bobst runs through this flow today.)

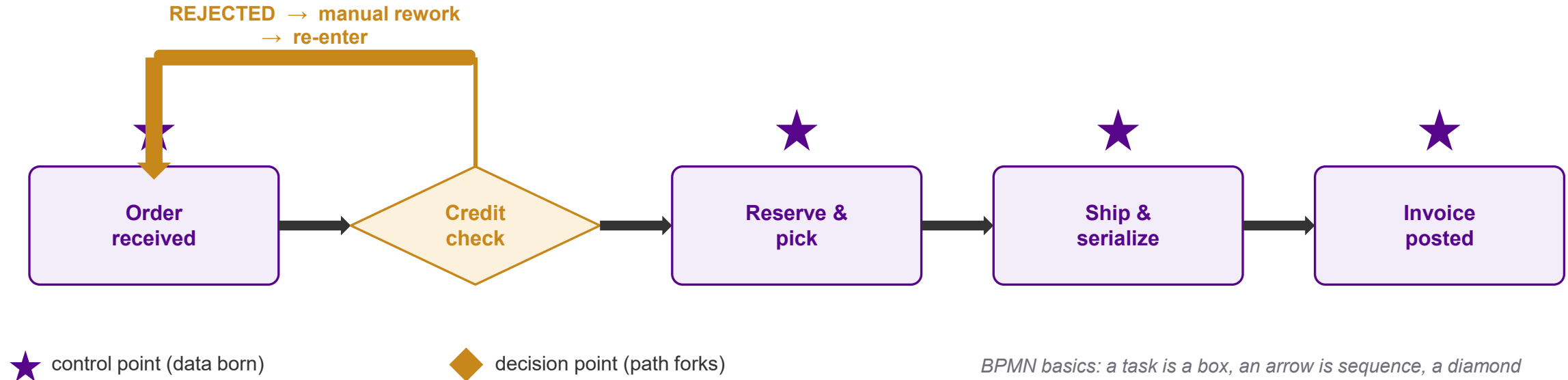
Learn their names now and watch for them all term. Almost every integration decision touches one of these two end-to-end flows. When you diagnose a company, you trace the flow first.

Control points: where the business commits

- Trace either flow and a few moments matter far more than the rest: a PO is issued, a serial or lot number generated, credit approved, an invoice posted.
- These are control points — the instants where the business commits and a new, authoritative fact comes into existence.
- They are where governance attaches, where an error is cheapest to catch, and — later in the course — where automation and AI agents plug in.
- So we read the process before the data. Data hangs on a process, not the other way round. Business first, data last.

A control point is where data is born. Find them first: in Lab 1 you'll mark them on your own process map. Map the business process and its control points first; fit the data to it last.

Decision points — and the rework loop where value leaks



BPMN basics: a task is a box, an arrow is sequence, a diamond gateway is a decision. That's all you need for Lab 1.

Rejection is the costly case. A rejected item doesn't stop — it's pushed back, fixed by hand, and re-enters only after rework. Each loop adds delay; each manual touch adds error. These loops are where value leaks fastest — and the richest targets in Lab 1.

They're also where the course's later question gets decided: which steps a rule-based robot can handle, and which need genuine human judgment.

TWO MINUTES — TURN TO YOUR NEIGHBOUR

Find one of each

1

In a process you know — buying something at work, getting reimbursed — name one control point (where a fact is committed).

2

Name one decision point in the same process — and say what happens to the work when it gets rejected.

A few pairs share out loud — this is the muscle Lab 1 exercises.

Two questions decide everything

Before it buys a thing, a company answers two separate questions about how it means to operate — and too often blurs them into one:

1 · Standardization

How identically should a process run across the company, whoever performs it and wherever? It drives out variability and lets you measure and improve.

The price: it limits local innovation, and the switch is expensive and bruising.

2 · Integration

How much should units share data — so a transaction flows end to end, and the company shows one face to the customer?

Its hardest part is the data: units must agree on common definitions before anything can flow.

Even the word “sale” isn’t safe. One unit dates a sale from the signed contract, another from cash received, a third from delivery. Until they agree, there is no single source of truth to share.

Four stances — and four real companies that took them



There is no best quadrant — only a stance that fits your strategy or fights it. Once chosen, the stance stops describing the company and starts driving it. Placing your company here is Board Memo Section 3.

The foundation that frees you

- The paradox: digitizing a core process makes that one process less flexible — it's in software now — and yet it makes the whole company more agile.
- Think of an athlete: fixed muscles and trained reflexes are exactly what let them react and improvise. A firm that has digitized what doesn't change is free to focus on what does.
- The foundation for execution becomes a platform for innovation precisely by making the routine boring.

~1/3

of firms had digitized their core — and enjoyed higher profitability, faster time to market, lower risk, and about a quarter lower IT costs.

The foundation is not a luxury that costs more — done right, it costs less. That's the case for treating the backbone as the main event, not an afterthought — the exact mistake this course exists to head off.

How to spot a company that lacks a foundation

- Different parts of the company give different answers to the same customer question.
- Meeting a new regulation is a massive, top-driven scramble.
- Every new initiative feels like starting from scratch.
- IT is the perpetual bottleneck — and people spend their days moving data from one system into another.

Learn to hear these in the wild — in a case, an interview, your own employer. Each is evidence the foundation is weak, whatever the software cost. Westerman's study of 400+ large firms found the winners aren't the biggest spenders — the “Digital Masters” run ~26% more profitable, by pairing digital capability with the leadership to drive change.

A backbone can work — and still fail to pay

- A company can select a backbone, implement it, survive go-live, and end up with a system that genuinely works — data flows, transactions post — and still find the business no better off.
- The promised savings don't materialise; the customer experience is unchanged; cash still moves at the old speed.
- The distance between a backbone that is deployed and one that delivers value is the leverage gap. Closing it is the real work of the next thirteen weeks.

>70%

of recently implemented ERP initiatives will, by 2027, fail to fully meet their original business goals — up to a quarter of them catastrophically.

Gartner

Read the number carefully. It isn't saying the software doesn't run. It's saying that most of the time, even a functioning system doesn't deliver what the business set out to get. That is the leverage gap, measured.

The same lesson, twenty-eight years apart

1998 — Davenport

Enterprise-system failures were business and organisational failures, not technical ones. The firms that won treated it as a redesign of who does what — they “stressed the enterprise, not the system.”

2026 — McKinsey

Money pours into AI agents while the ERP foundation they depend on is neglected — so only a minority of companies report enterprise-level profit from AI. An agent on a fragmented backbone just acts on bad data faster.

The floor still has to be solid before anything stands on it — autonomous or not. That's why this course opens here. The leverage gap you just met is Davenport's warning, resurfaced with AI in the role software once played.

BEHIND TODAY'S FLOOR Davenport, HBR (1998, NYU Libraries) · McKinsey, “Bridging the great AI agent and ERP divide” (2026) · video: Goss, “SAP ERP, ECC and S/4HANA” (Eye on Tech) · SAP, “What is ERP”

A first look at friction: the Four Lenses

Frustration

Where the work irritates the people doing it or the customer receiving it — a tell for hidden rework and workarounds.

Time

Where the process waits, queues, or doubles back — elapsed time far exceeding actual work time.

Cost

Where effort, rework, or error-correction consumes money out of proportion to the value the step adds.

Quality

Where errors and defects are produced — then caught and fixed somewhere downstream.

A lens looks at a process, not a database. You find the leak in the work, then ask what data would prove it. We preview them today; the full diagnostic is Week 2, and you first apply them in the Week 2 pre-work and Lab 1.

NYU Bobst Library: the problem

Primary case · Ferguson (2025), open access

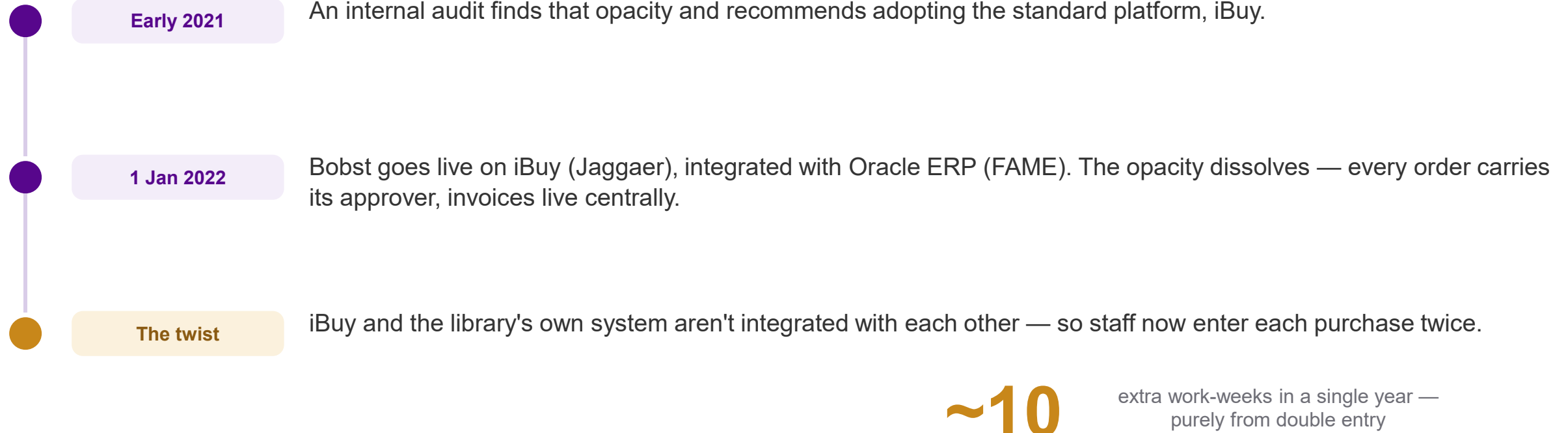
- 2014–2021: Bobst paid most collections invoices through a homegrown workflow that bypassed iBuy, the university's standard procurement platform.
- Staff ordered and received in the library's own system; a weekly job pushed invoice data straight to Accounts Payable.
- To the people using it, it felt efficient.

To everyone outside the library, it was opaque:

- no record of who authorised the spending
- no metadata on what was purchased
- no copies of the invoices retained centrally

A small case by design — the point is clarity, not scale: a Procure-to-Pay flow you can hold in your hand.

Bobst: solving one gap opened another



This is a missing foundation's textbook symptom. People moving data from one system into another — re-created at a new seam by a technically successful migration. Integration is a gap you keep closing, not a finish line.

Bobst: the lesson, and the thread it lands on

- The backbone did its job right the first time: adopting the standard platform restored a single source of truth and pulled a rogue process back into the central flow.
- And yet it re-created the very symptom of a missing foundation — daily data re-entry — at a new seam.
- Integration is not a state you arrive at; it's a gap you keep closing, and closing one can open the next.

The thread this lands on:

Cash Management (primary)

Operations (overlay)

— the currency in which the double-entry cost is actually paid.

This is exactly the diagnosis your team runs in the breakout. Trace the flow, mark where data is born and where it's re-keyed, and name what the seam costs — on a named thread. Bobst is your worked example.

Lidl and SAP: half a billion euros, no value

Supporting case · Computer Weekly + Handelsblatt

€500M

reported spend

7 yrs

2011 → 2018

0

value delivered — reverted to legacy

- Not a story about bad software. Lidl values inventory at the price it paid; standard SAP for Retail is built around the retail price.
- Faced with the mismatch, Lidl bent the software to its processes rather than its processes to the software — and the customization compounded until the program collapsed.

Customize vs configure — the single most repeated ERP error. A configured system stays close to the vendor standard and survives upgrades; a heavily customized one drifts until routine updates break it. The backbone was deliverable but never adoptable.

How to read a backbone — six steps

- 1 Trace the flows**
Follow the O2C or P2P that matters most. Where is each fact born, rekeyed, or stalled?
- 2 Mark the points**
Control points (data born) and decision points (path forks). Watch the reject-and-rework loops.
- 3 Classify the model**
Ross & Weill's two questions place the company in one of the four stances.
- 4 Run the Four Lenses**
Frustration, Time, Cost, Quality — where the work leaks value.
- 5 Listen for warnings**
Different answers to one question; every initiative from scratch; IT the bottleneck.
- 6 Name the leverage gap**
Does the backbone work but not pay? Where, on which thread — Customer, Operations, or Cash?

Run those six steps and you've read a backbone. It's what the cases showed — and exactly what your team starts doing in the breakout, and repeats in Lab 1 and the capstone.

Find your team

Teams are already assigned. Grab water, sit together, and get ready — when you sit down, you start your capstone.

TEAM 1

- Paola Munoz · IE
- Qiheng Li · EE
- Darsh Desai · IE
- Hashir Muzaffar · CompE
- Jeongil Seo · IE

TEAM 2

- Vania Salsabila · IE
- Quincy Oldland · CS
- Tanish Vardhini · CompE
- Steven Hsu · IE
- Dyson Ye · IE

TEAM 3

- Emma Hwang · IE
- Jasper von Selchow · MOT
- Siddharth Sameer · CompE
- Wirasana Andiputra · IE
- Gunnur Magnadóttir · IE

TEAM 4

- Yu Hua Yu · IE
- Rahul Nair · IE
- Salauat Kakimzhanov · CompE
- Alan Sarieddine · IE
- Xinyi Ju · IE

Full roster is posted on Brightspace. Not here today? Join any team of four.

Meet, then pick your company

First 5 minutes — introduce yourselves

You'll work with these people all term. Go around the table — about a minute each:

- Professional — your background, and a system you've actually used (SAP, Salesforce, QuickBooks...).
- Personal — what brought you to this course.
- A fun fact — something no one would guess.

Then — pick a candidate company

Choose one together. Good picks are:

- One a teammate knows from the inside — a current/former employer or internship.
- Real and publicly researchable — a website, filings, job ads to read.
- Big enough to have interesting integration seams, small enough to hold in your head.

This is a practice run on step 1 — you can change the company later. Getting a real one on the table now is what makes the rest of the term concrete. Don't over-think it; pick, and start reading its backbone.

Read your company's backbone

Run these six moves on the company you chose. Where you can't confirm a fact, make a labelled assumption. Put one person on a shared doc; be ready to share in 60 seconds.

- 1 Get the context** What do they do, in what industry, at what scale (revenue, sites, headcount), and how do they make money? Who are their main customers — and which thread matters most: Customer, Operations, or Cash?
- 2 Name the backbone** One ERP, a stitched SaaS stack, or spreadsheets? Where is its single source of truth — or where is it missing?
- 3 Place it** Which of the four archetypes — and cloud/on-prem, generalist/vertical, tier — read off job ads and the website.
- 4 Trace one flow** Sketch an O2C or P2P flow; mark one control point (data born) and one decision point (path forks).
- 5 Classify the model** Ross & Weill's two questions place it in one of the four stances (reference is coming up).
- 6 Name the leverage gap** One sentence: the backbone works, but where does value fail to appear — Customer, Operations, or Cash?

YOU LEAVE WITH

a short company context, a backbone read, its operating model, and a one-sentence leverage gap on a named thread. This is the seed of your backbone-context note, due for Week 2.

A few teams share; the instructor sharpens. Move fast — a first pass beats a perfect one.

Chipotle Mexican Grill — context + backbone

1 · Context RESEARCHED

Fast-casual Mexican restaurant operator. ~4,050 restaurants, almost all company-operated — no traditional franchising. FY2025 revenue ~\$11.9B; digital ~37% of food & beverage sales (app, web, third-party delivery), plus Chipotlane drive-thru pickup.

Primary threads: Operations & Supply Chain (kitchen throughput, fresh-ingredient supply) and Customer (fulfilment across in-store, app, delivery).

2 · Backbone STACK ASSUMED

POS + digital ordering (app/web) + delivery + inventory/supply chain + loyalty on one shared data model. Single source of truth = the order-and-inventory record every channel writes to.

3 · Place it TIER/SCOPE ASSUMED

Scale resembles the large-enterprise archetype. Likely a cloud, food-service/retail-vertical, Tier I backbone — confirm from job ads (“SAP”, “Workday”, POS vendor).

Green = researched from public filings (Chipotle FY2025 results). Gold = a labelled assumption to validate — exactly the mix your output should show.

Chipotle — flow, model, and the leverage gap

4 · Trace one flow

A digital order (app / web / delivery) is routed to a restaurant, prepared on the make-line, and handed off in-store, at a Chipotle, or to a courier — then payment is captured.

★ **Control point:** order committed — payment authorized, the order record is born.

◆ **Decision point:** at peak, accept the order now or delay it — capacity vs demand.

5 · Operating model: **Unification**

High standardization (identical menu and make-line in every unit) and high integration (one backbone). Owning every restaurant makes standardizing cheap.

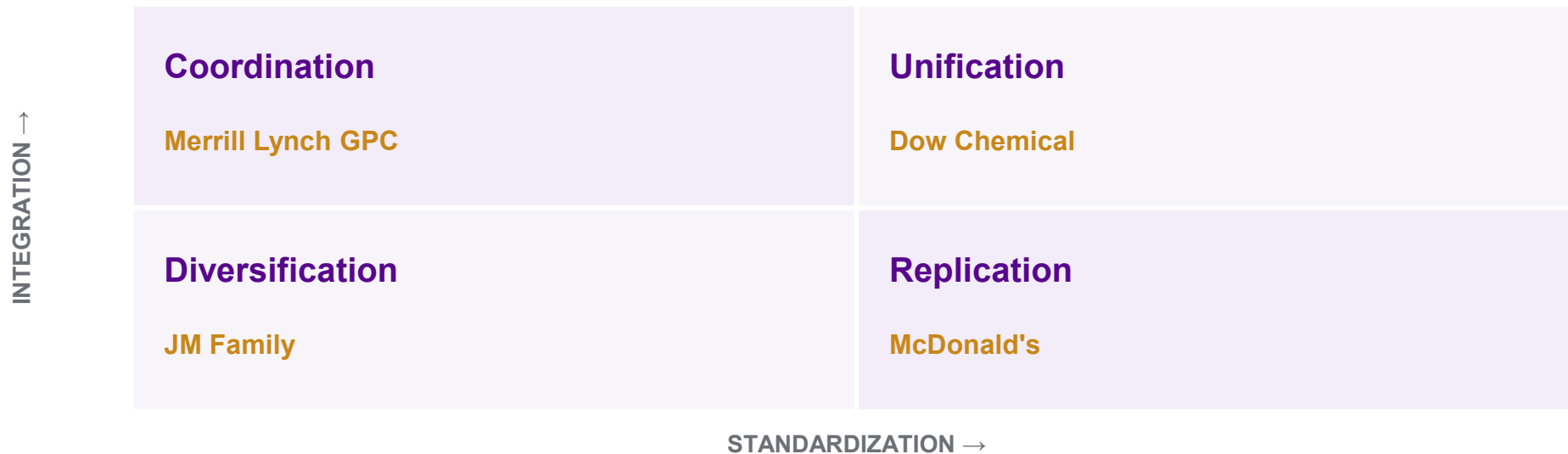
6 · Leverage gap **HYPOTHESIS**

Digital grew to ~37% of sales, but if that order stream isn't tightly synced to real-time kitchen capacity and inventory, peak digital orders strain the make-line — waits, stockouts, remakes. The backbone works, yet value leaks on Operations & Customer.

Your one-sentence output: “Chipotle's backbone runs a standardized, company-owned operation well, but its fast-growing digital demand may outrun real-time kitchen and inventory sync — a throughput leak on the Operations and Customer threads.” A gap stated as a hypothesis, ready to validate.

The two questions, and the four stances

- Standardization — how identically should the process run everywhere?
- Integration — how much should units share data, end to end and to the customer?



No best quadrant — only a stance that fits your strategy or fights it.

What a good read sounds like

- The single source of truth looks different at every scale — one SAP instance, three stitched SaaS tools, a shared spreadsheet held together by a person — but the failure mode is identical: without one authoritative record, reconciliation becomes a tax.
- There's no best quadrant. A stance that fits one company's strategy would fight another's.
- Naming the thread — Customer, Operations, or Cash — is already the start of your Board Memo Section 2.

Carry your placement forward. The archetype, the operating model, and the one-sentence leverage gap you named are the seed of the backbone-context note due for Week 2 — and every later recommendation inherits from it.

Confirm your company with the instructor by the end of this week — a real organisation a teammate can learn about.

Why a working backbone still fails to pay

Week 2 — Failure to Leverage + BPR as Remedy

- We end on the leverage gap because Week 2 dissects it — the difference between a system that works and one that pays.
- The remedy isn't better technology; it's redesigning the process before you automate it (Business Process Re-engineering).
- You'll meet two failures a generation apart — one a global brand, one a public authority — that fell into the identical trap.

Read the Week 2 note this week — you have a full week's lead. From here, one note before each class.

Before Week 2: read, start your memo, set up

1 · Read

The Week 2 study note — “Failure to Leverage + BPR as Remedy” — before next class. This is the rhythm from now on.

2 · Start the memo

Your backbone-context note — stack, key processes, archetype, primary thread. That's Board Memo Section 2, and it fills storyboard slides S1 and S3.

3 · Set up

Create your free, browser-based Default-stack accounts (Airtable or Google Sheets + Gemini) during the first week. Your team declares its stack path by the end of Week 3 — see the next slide.

You leave today with a team, a company in mind, and the storyboard. Confirm the company with the instructor by the end of this week — a real organisation, with synthetic data only.

Three paths — the Default is enough

The labs are outcome-defined: you meet the intent and deliverables; how you build is largely your choice. Pick a path and declare it by the end of Week 3.

DEFAULT

Guaranteed · no approval · full marks

Free, browser-based, no code: Miro or Lucidchart (map), Airtable or Google Sheets + Gemini (data), Make (automate), Relevance AI (agent).

ENTERPRISE — AWS

Partner-supported · onboarding + notify

Our AWS partner's path: Bedrock playground, Step Functions + EventBridge, Bedrock Agents / AgentCore.

CUSTOM

Faculty approval · declare by Week 3

Your own mix — Default, AWS, or other platforms — where it genuinely serves your solution. Just declare and justify the choice.

You're graded on business outcomes, governance, and solution design — never platform choice or complexity. The Default earns full marks and is never a disadvantage. Choose AWS or Custom only if it genuinely serves the problem — not to look sophisticated.

Where we are — and the compact

- A backbone is a single source of truth that also enforces the process — table stakes, not the prize.
- You can read any backbone with one method: trace the flows, mark the points, classify the model, run the lenses, hear the warnings, name the gap.
- A backbone that works can still fail to pay. That gap — the leverage gap — is the whole course.

One thread we carry all module: what's owed to people when a system standardises a role away? When integration eliminates work, who decides what's automated first, and what process is due? Hold the question; we return to it as the automation gets real.

DUE FOR WEEK 2 Read the W2 note · start the backbone-context note (S1/S3, Board Memo §2) · confirm your company.

Next week

Why a working backbone fails to pay

Week 2 — Failure to Leverage + BPR as Remedy. Read the note; bring your backbone-context note and your leverage gap.